



DEPARTMENT OF MATHEMATICS  
**ACADEMIC YEAR: 2023-24 EVEN SEMESTER (BMAE201)**  
**QUESTION BANK**  
**MODULE -3**

1. Find the Laplace transform of (i)  $te^{-t}\sin 4t$  (ii)  $\frac{(1-\cos at)}{t}$
2. Find the Laplace transform of (i)  $e^{-3t}(2\cos 5t - 3\sin 5t)$  (ii)  $\frac{(\cos at - \cos bt)}{t}$
3. Find the Laplace transform of (i)  $e^{-4t}(2\cos 6t - 3\sin 5t)$  (ii)  $\frac{(\cos 2t - \cos 3t)}{t}$
4. Find the Laplace transform of the triangular wave function  $f(t) = \begin{cases} t & \text{if } 0 \leq t \leq a \\ 2a - t & \text{if } a \leq t \leq 2a \end{cases}$
5. Find the Laplace transform of the square wave function of period  $2a$ , defined by  $f(t) = \begin{cases} E & \text{if } 0 \leq t \leq T/2 \\ -E & \text{if } T/2 \leq t \leq T \end{cases}$  show that  $L\{f(t)\} = \frac{E}{s} \tanh\left(\frac{sT}{4}\right)$
6. Find the Laplace transform of the square wave function of period  $2a$ , defined by  $f(t) = \begin{cases} k & \text{if } 0 \leq t \leq a \\ -k & \text{if } a \leq t \leq 2a \end{cases}$
7. Express the following function in terms of unit step function and hence find its Laplace transform of  $f(t) = \begin{cases} t^2 & \text{if } 0 < t < 2 \\ 4t & \text{if } 2 < t < 4 \\ 8 & \text{if } t > 4 \end{cases}$
8. Express  $f(t) = \begin{cases} \cos t & 0 < t < \pi \\ \cos 2t & \pi < t < 2\pi \\ \cos 3t & \text{if } t > \pi \end{cases}$  in terms of the Heaviside unit step function and hence find  $\{f(t)\}$ .
9. Express  $f(t) = \begin{cases} \sin t & 0 < t < \pi \\ \sin 2t & \pi < t < 2\pi \\ \sin 3t & \text{if } t > \pi \end{cases}$  in terms of the Heaviside unit step function and hence find  $\{f(t)\}$ .
10. Find the inverse Laplace transform of  
(i)  $\frac{s}{s^2+2s+3}$  (ii)  $\frac{1}{(s+2)^2}$  (iii)  $\frac{2s^2-6s+5}{s^3-6s^2+11s-6}$  (iv)  $\frac{1}{s(s+1)(s+2)(s+3)}$
11. Using the convolution theorem, find the inverse Laplace transform of  $\frac{1}{s^3(s^2+1)}$
12. Using the convolution theorem, find the inverse Laplace transform of  $\frac{s}{(s^2+a^2)^2}$
13. Solve by Laplace transform method:  $y'' + 4y' + 3y = e^{-t}$  with  $y(0) = y'(0) = 1$
14. Solve the differential equation by using the Laplace transform method  
 $\frac{d^2y}{dt^2} + y = \sin 2t$  with  $y(0) = y'(0) = 0$



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